

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 18 June 2026

Team ID: VIP-C2-Book-A-Doctor-MERN

Project Name: Book A Doctor – Online Doctor Appointment Booking System

Technical Architecture

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| Patients / Doctors |

| / Admin |

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| React.js Frontend |

| (User Interface) |

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| Node.js + Express.js |

| REST API Backend |

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| MongoDB Atlas |

| Cloud Database |

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Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web application interface for patients, doctors, and admins	React.js, HTML5, CSS3, JavaScript
2	Application Logic-1	User authentication and authorization	Node.js, Express.js
3	Application Logic-2	Appointment booking and management	Node.js, Express.js
4	Application Logic-3	Notification and doctor approval workflow	Node.js, Express.js
5	Database	Stores users, doctors, appointments, and notifications	MongoDB
6	Cloud Database	Cloud-hosted database service	MongoDB Atlas
7	File Storage	Stores application assets and documents	Local Storage / Cloud Storage
8	External API-1	REST API communication between frontend and backend	Axios API Calls
9	External API-2	Authentication token verification	JWT
10	Machine Learning Model	Not Applicable in current version	N/A
11	Infrastructure (Cloud)	Application deployment and hosting	Vercel (Frontend), Render (Backend)

Table-2 : Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for application development	React.js, Node.js, Express.js
2	Security Implementations	Password encryption, authentication, protected routes	JWT, bcryptjs

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|---|-----------------------|---|
| 3 | Scalable Architecture | Modular MERN architecture supports future expansion |
| | MERN Stack | |
| 4 | Availability | Cloud deployment ensures accessibility from anywhere Vercel, Render, MongoDB Atlas |
| 5 | Performance | Fast API communication and optimized frontend rendering React.js, Express.js, Axios |

Architecture Benefits

- Secure user authentication and authorization.
- Cloud-based database access.
- Fast and responsive user experience.
- Easy maintenance and scalability.
- Supports future enhancements such as telemedicine and online payments.

Conclusion

The Book A Doctor platform is built using a modern MERN Stack architecture that provides scalability, security, reliability, and high performance. The selected technology stack ensures efficient appointment management and supports future growth of the application.